

**MALAVIYA NATIONAL INSTITUTE OF TECHNOLOGY JAIPUR**  
**DEPARTMENT OF CHEMISTRY**

Name of the Student \_\_\_\_\_ ID No. of the Student \_\_\_\_\_

Chemistry Laboratory Quiz

19/11/2025

**Marks: 20**

**Time: 60 minutes**

- Q1. 50 ml of a hard water sample consumed 15 mL of 0.01M-EDTA before boiling and 5 mL of the same EDTA after boiling. Calculate the degree of total hardness, permanent hardness and temporary hardness. What is the role of Mg-EDTA in the estimation of hardness of water using complexometric titration? [5 marks]
- Q2. Determine the equivalent mass of sodium thiosulphate ( $\text{Na}_2\text{S}_2\text{O}_3 \cdot 5\text{H}_2\text{O}$ ) in the reaction,  $2\text{Na}_2\text{S}_2\text{O}_3 + \text{I}_2 \rightarrow 2\text{NaI} + \text{Na}_2\text{S}_4\text{O}_6$ . The given molecular weight of  $\text{Na}_2\text{S}_2\text{O}_3 \cdot 5\text{H}_2\text{O}$  is 248.17 g. [3 marks]
- Q3. 3.55 g of bleaching powder when treated with acetic acid and excess of KI liberated iodine which required 60 mL of 0.5 N sodium thiosulphate solution. Calculate the percentage of available chlorine in the sample of bleaching powder. [5 marks]
- Q4. In iodometric titration why starch is not used at the beginning of the experiment? [3 marks]
- Q5. Write the chemical reactions occurring in the Modified Winkler method, including azide removal of nitrite. A sample treated using the modified Winkler procedure required 7.2 mL of 0.05 N  $\text{Na}_2\text{S}_2\text{O}_3$ . If the sample volume was 250 mL, calculate the DO in  $\text{mg/L}$ . [4 marks]